Class 9 Mathematics — Exercise 4.2

Basic Concept

Formulas

1. $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$

2. $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$

3. $(a+b)^3 = a^3 + b^3 + 3ab(a+b) = a^3 + b^3 + 3a^2b + 3ab^2$

4. $(a-b)^3 = a^3 - b^3 - 3ab(a-b) = a^3 - b^3 - 3a^2b + 3ab^2$

Method for Q13 to Q20

When an expression has a **common part**, put that part equal to a ("Let … $= a$"), factorize the simple quadratic in a by splitting the middle term, then **put the value of** a back.

For a product of four brackets, first **pair the brackets** so that both pairs give the same common part.

Question 1

$$x^3 - 125$$

$$= (x)^3 - (5)^3 \qquad (\because a^3 - b^3 = (a-b)(a^2 + ab + b^2))$$
$$= (x-5)[(x)^2 + (x)(5) + (5)^2]$$
$$= \boxed{(x-5)(x^2 + 5x + 25)} \text{ Ans}$$

Question 2

$$8x^3 + 1$$

$$= (2x)^3 + (1)^3 \qquad (\because a^3 + b^3 = (a+b)(a^2 - ab + b^2))$$
$$= (2x+1)[(2x)^2 - (2x)(1) + (1)^2]$$
$$= \boxed{(2x+1)(4x^2 - 2x + 1)} \text{ Ans}$$

Question 3

$$3p^3q^3 - 81x^3$$

$$= 3(p^3q^3 - 27x^3)$$
$$= 3[(pq)^3 - (3x)^3] \qquad (\because a^3 - b^3 = (a-b)(a^2+ab+b^2))$$
$$= 3(pq - 3x)[(pq)^2 + (pq)(3x) + (3x)^2]$$
$$= \boxed{3(pq - 3x)(p^2q^2 + 3pqx + 9x^2)} \text{ Ans}$$

Question 4

$$27 + 512x^3$$

$$= (3)^3 + (8x)^3 \qquad (\because a^3 + b^3 = (a+b)(a^2 - ab + b^2))$$
$$= (3 + 8x)[(3)^2 - (3)(8x) + (8x)^2]$$
$$= \boxed{(3 + 8x)(9 - 24x + 64x^2)} \text{ Ans}$$

Question 5

$$t^6 - 64$$

$$= (t^3)^2 - (8)^2 \qquad (\because a^2 - b^2 = (a+b)(a-b))$$
$$= (t^3 + 8)(t^3 - 8)$$
$$= [(t)^3 + (2)^3][(t)^3 - (2)^3]$$
$$= (t+2)[(t)^2 - (t)(2) + (2)^2]\ (t-2)[(t)^2 + (t)(2) + (2)^2]$$
$$= \boxed{(t+2)(t^2 - 2t + 4)(t-2)(t^2 + 2t + 4)} \text{ Ans}$$

Question 6

$$x^6 + y^6$$

$$= (x^2)^3 + (y^2)^3 \qquad (\because a^3 + b^3 = (a+b)(a^2 - ab + b^2))$$
$$= (x^2 + y^2)[(x^2)^2 - (x^2)(y^2) + (y^2)^2]$$
$$= \boxed{(x^2 + y^2)(x^4 - x^2y^2 + y^4)} \text{ Ans}$$

Question 7

$$(2 - x)^3 + (y - 2)^3$$

Here $a = 2 - x$ and $b = y - 2$. $\qquad (\because a^3 + b^3 = (a+b)(a^2 - ab + b^2))$

$$= (2 - x + y - 2)[(2 - x)^2 - (2 - x)(y - 2) + (y - 2)^2]$$
$$= (y - x)[(2)^2 + (x)^2 - 2(2)(x) - (2y - 4 - xy + 2x) + (y)^2 + (2)^2 - 2(y)(2)]$$
$$= (y - x)(4 + x^2 - 4x - 2y + 4 + xy - 2x + y^2 + 4 - 4y)$$
$$= \boxed{(y - x)(x^2 + y^2 + xy - 6x - 6y + 12)} \text{ Ans}$$

Question 8

$$64(x+y)^3 - z^3$$

$$= 4^3(x+y)^3 - z^3$$
$$= [4(x+y)]^3 - (z)^3 \qquad (\because a^3 - b^3 = (a-b)(a^2+ab+b^2))$$
$$= (4x+4y-z)[(4x+4y)^2 + (4x+4y)(z) + (z)^2]$$
$$= \boxed{(4x+4y-z)(16x^2+16y^2+32xy+4xz+4yz+z^2)} \text{ Ans}$$

Question 9

$$27p^3 + 144pq^2 - 108p^2q - 64q^3$$

$(\because a^3 - 3a^2b + 3ab^2 - b^3 = (a-b)^3)$

$$= 27p^3 - 108p^2q + 144pq^2 - 64q^3$$
$$= (3p)^3 - 3(3p)^2(4q) + 3(3p)(4q)^2 - (4q)^3$$
$$= \boxed{(3p-4q)^3} \text{ Ans}$$

Question 10

$$8p^3 + q^3 + 12p^2q + 6pq^2$$

$(\because a^3 + b^3 + 3a^2b + 3ab^2 = (a+b)^3)$

$$= (2p)^3 + (q)^3 + 3(2p)^2(q) + 3(2p)(q)^2$$
$$= \boxed{(2p+q)^3} \text{ Ans}$$

Question 11

$$125x^3 - y^3 - 75x^2y + 15xy^2$$

$(\because a^3 - 3a^2b + 3ab^2 - b^3 = (a-b)^3)$

$$= (5x)^3 - 3(5x)^2(y) + 3(5x)(y)^2 - (y)^3$$
$$= \boxed{(5x-y)^3} \text{ Ans}$$

Question 12

$$p^3 - 9p^2q + 27pq^2 - 27q^3$$

$$= (p)^3 - 3(p)^2(3q) + 3(p)(3q)^2 - (3q)^3$$
$$= \boxed{(p-3q)^3} \text{ Ans}$$

Question 13

$$(2x^2 - 3x + 6)(2x^2 - 3x) - 55$$

Let $2x^2 - 3x = a$

$$= (a + 6)(a) - 55$$
$$= a^2 + 6a - 55$$

$-55 \rightarrow 11$ and $-5 \quad (11 - 5 = 6)$

$$= a^2 + 11a - 5a - 55$$
$$= a(a + 11) - 5(a + 11)$$
$$= (a + 11)(a - 5)$$

Put value of a:

$$= (2x^2 - 3x + 11)(2x^2 - 3x - 5)$$

($2x^2 - 3x + 11$ cannot be factorized further. For $2x^2 - 3x - 5$: $-10 \rightarrow -5$ and 2)

$$= (2x^2 - 3x + 11)[2x^2 + 2x - 5x - 5]$$
$$= (2x^2 - 3x + 11)[2x(x + 1) - 5(x + 1)]$$
$$= \boxed{(2x^2 - 3x + 11)(x + 1)(2x - 5)} \text{ Ans}$$

Question 14

$$(y^2 + 2y - 3)(y^2 + 2y + 11) + 48$$

Let $y^2 + 2y = a$

$$= (a - 3)(a + 11) + 48$$
$$= a(a + 11) - 3(a + 11) + 48$$
$$= a^2 + 11a - 3a - 33 + 48$$
$$= a^2 + 8a + 15$$

$15 \rightarrow 5$ and $3 \quad (5 + 3 = 8)$

$$= a^2 + 5a + 3a + 15$$
$$= a(a + 5) + 3(a + 5)$$
$$= (a + 5)(a + 3)$$

Put value of a:

$$= \boxed{(y^2 + 2y + 3)(y^2 + 2y + 5)} \text{ Ans}$$

Question 15

$$y(y-1)(y-3)(y-4)+2$$

Pairing: $(y+0)+(y-4)$ gives -4, and $(y-1)+(y-3)$ gives -4.

$$\begin{aligned}
&= [y(y-4)][(y-1)(y-3)]+2 \\
&= (y^2-4y)(y^2-3y-y+3)+2 \\
&= (y^2-4y)(y^2-4y+3)+2
\end{aligned}$$

Let $y^2-4y=a$

$$\begin{aligned}
&= a(a+3)+2 \\
&= a^2+3a+2
\end{aligned}$$

$2 \rightarrow 2$ and $1 \quad (2+1=3)$

$$\begin{aligned}
&= a^2+2a+a+2 \\
&= a(a+2)+1(a+2) \\
&= (a+2)(a+1)
\end{aligned}$$

Put value of a:

$$= \boxed{(y^2-4y+2)(y^2-4y+1)} \text{ Ans}$$

Question 16

$$(k+2)(k-3)(k+5)(k+10)+375$$

Pairing: $2+5=7$ and $-3+10=7$

$$\begin{aligned}
&= (k+2)(k+5)(k-3)(k+10)+375 \\
&= [k(k+5)+2(k+5)][k(k+10)-3(k+10)]+375 \\
&= (k^2+7k+10)(k^2+7k-30)+375
\end{aligned}$$

Let $k^2+7k=a$

$$\begin{aligned}
&= (a+10)(a-30)+375 \\
&= a(a-30)+10(a-30)+375 \\
&= a^2-30a+10a-300+375 \\
&= a^2-20a+75
\end{aligned}$$

$75 \rightarrow -15$ and $-5 \quad (-15-5=-20)$

$$\begin{aligned}
&= a^2-15a-5a+75 \\
&= a(a-15)-5(a-15)
\end{aligned}$$

$$= (a-15)(a-5)$$

Put value of a:

$$= \boxed{(k^2+7k-15)(k^2+7k-5)} \text{ Ans}$$

Question 17

$$(x-5)(x-6)(x+3)(x+2)+12$$

Pairing: $-5+2=-3$ and $-6+3=-3$

$$= (x-5)(x+2)(x-6)(x+3)+12$$
$$= [x(x+2)-5(x+2)][x(x+3)-6(x+3)]+12$$
$$= (x^2-3x-10)(x^2-3x-18)+12$$

Let $x^2-3x=a$

$$= (a-10)(a-18)+12$$
$$= a(a-18)-10(a-18)+12$$
$$= a^2-18a-10a+180+12$$
$$= a^2-28a+192$$

$192 \rightarrow -16$ and $-12 \quad (-16-12=-28)$

$$= a^2-16a-12a+192$$
$$= a(a-16)-12(a-16)$$
$$= (a-16)(a-12)$$

Put value of a:

$$= \boxed{(x^2-3x-16)(x^2-3x-12)} \text{ Ans}$$

Question 18

$$(x+1)(x+2)(x-3)(x-6)-21x^2$$

Pairing: $1\times(-6)=-6$ and $2\times(-3)=-6$

$$= (x+1)(x-6)(x+2)(x-3)-21x^2$$
$$= [x(x-6)+1(x-6)][x(x-3)+2(x-3)]-21x^2$$
$$= (x^2-6x+x-6)(x^2-3x+2x-6)-21x^2$$
$$= (x^2-5x-6)(x^2-x-6)-21x^2$$
$$= (x^2-6-5x)(x^2-6-x)-21x^2$$

Let $x^2-6=a$

$$= (a-5x)(a-x) - 21x^2$$
$$= a(a-x) - 5x(a-x) - 21x^2$$
$$= a^2 - ax - 5ax + 5x^2 - 21x^2$$
$$= a^2 - 6ax - 16x^2$$

$-16 \rightarrow -8$ and $+2 \quad (-8+2=-6)$

$$= a^2 - 8ax + 2ax - 16x^2$$
$$= a(a-8x) + 2x(a-8x)$$
$$= (a-8x)(a+2x)$$

Put value of a:

$$= \boxed{(x^2-6-8x)(x^2-6+2x)} \text{ Ans}$$

i.e. $(x^2-8x-6)(x^2+2x-6)$

Question 19

$$(x-2)(x-6)(x-3)(x-4) - 2x^2$$

Pairing: $(-2)(-6) = 12$ and $(-3)(-4) = 12$

$$= [x(x-6) - 2(x-6)][x(x-4) - 3(x-4)] - 2x^2$$
$$= (x^2 - 6x - 2x + 12)(x^2 - 4x - 3x + 12) - 2x^2$$
$$= (x^2 - 8x + 12)(x^2 - 7x + 12) - 2x^2$$
$$= (x^2 + 12 - 8x)(x^2 + 12 - 7x) - 2x^2$$

Let $x^2 + 12 = a$

$$= (a-8x)(a-7x) - 2x^2$$
$$= a(a-7x) - 8x(a-7x) - 2x^2$$
$$= a^2 - 7ax - 8ax + 56x^2 - 2x^2$$
$$= a^2 - 15ax + 54x^2$$

$54 \rightarrow -9$ and $-6 \quad (-9-6=-15)$

$$= a^2 - 9ax - 6ax + 54x^2$$
$$= a(a-9x) - 6x(a-9x)$$
$$= (a-9x)(a-6x)$$

Put value of a:

$$= \boxed{(x^2+12-9x)(x^2+12-6x)} \text{ Ans}$$

i.e. $(x^2-9x+12)(x^2-6x+12)$

Question 20

$$(5-x)(2+x)(10-x)(1+x)-7x^2$$

Pairing: $5\times 2=10$ and $10\times 1=10$

$$\begin{aligned}
&= [5(2+x)-x(2+x)][10(1+x)-x(1+x)]-7x^2\\
&= (10+5x-2x-x^2)(10+10x-x-x^2)-7x^2\\
&= (10+3x-x^2)(10+9x-x^2)-7x^2\\
&= (10-x^2+3x)(10-x^2+9x)-7x^2
\end{aligned}$$

Let $10-x^2=a$

$$\begin{aligned}
&= (a+3x)(a+9x)-7x^2\\
&= a(a+9x)+3x(a+9x)-7x^2\\
&= a^2+9ax+3ax+27x^2-7x^2\\
&= a^2+12ax+20x^2
\end{aligned}$$

$20 \rightarrow 10$ and $2 \quad (10+2=12)$

$$\begin{aligned}
&= a^2+10ax+2ax+20x^2\\
&= a(a+10x)+2x(a+10x)\\
&= (a+10x)(a+2x)
\end{aligned}$$

Put value of a:

$$= \boxed{(10+10x-x^2)(10+2x-x^2)} \text{ Ans}$$

Question 21

The expression a^6+729 can be written in two ways as: (a) sum of two squares (b) sum of two cubes. Which one will be used for factoring it and why? Also factorize the given expression.

$$a^6+729$$

(a) $(a^3)^2+(27)^2$ (b) $(a^2)^3+(9)^3$

We **cannot** factorize part (a), because there is **no formula** for the sum of two squares (a^2+b^2). But we **can** factorize part (b) by using the formula $a^3+b^3=(a+b)(a^2-ab+b^2)$:

$$\begin{aligned}
a^6+729 &= (a^2)^3+(9)^3\\
&= (a^2+9)[(a^2)^2-(a^2)(9)+(9)^2]\\
&= \boxed{(a^2+9)(a^4-9a^2+81)} \text{ Ans}
\end{aligned}$$

Question 22

Express $8 + 12t + 6t^2 + t^3$ as the product of three factors. Is each factor a binomial or a trinomial?

$$8 + 12t + 6t^2 + t^3$$

$(\because a^3 + 3a^2b + 3ab^2 + b^3 = (a + b)^3)$

$$\begin{aligned} &= (2)^3 + 3(2)^2(t) + 3(2)(t)^2 + (t)^3 \\ &= (2 + t)^3 \\ &= (t + 2)^3 \\ &= \boxed{(t + 2)(t + 2)(t + 2)} \end{aligned}$$

Each factor is a binomial.